# Math 110 — Week 7

## Diagonalizability and the minimal polynomial (Notes Ch. 12–13, LADR 5C, 5D, 5B, HW7)

## §1 Facts used from earlier weeks

$V$ is finite-dimensional over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$), $n = \dim V$. Shorthand: "$S$ kills $v$" means $Sv = 0$, and "$q$ kills $T$" means $q(T) = 0$. From Week 6: eigenvalues, eigenvectors and eigenspaces $E(\lambda, T) = \operatorname{null}(T - \lambda I)$, and invariant subspaces.

(T1) **Rank–nullity.** For linear $T\colon V \to W$: $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$.

**Why.**

1. Extend a basis $u_1, \dots, u_k$ of $\operatorname{null} T$ to a basis $u_1, \dots, u_k, v_1, \dots, v_r$ of $V$. It suffices to show $Tv_1, \dots, Tv_r$ is a basis of $\operatorname{range} T$.
2. Spanning: $T$ of any vector is a combination of the $Tu_i$ (which are 0) and the $Tv_j$.
3. Independent: if $\sum c_j T v_j = 0$, then $\sum c_j v_j \in \operatorname{null} T$ is a combination of the $u_i$, and independence of the basis of $V$ gives all $c_j = 0$.

(T2) **For operators, injective $\iff$ surjective $\iff$ invertible.**

**Why.** $\operatorname{null} T = 0 \iff \dim \operatorname{range} T = n$ (T1) $\iff \operatorname{range} T = V$ (T3).

(T3) **Dimension shortcuts.**

(a) An independent list of $n$ vectors is a basis.

(b) A subspace of dimension $n$ is all of $V$.

(c) If $U \cap W = \{0\}$ and $\dim U + \dim W = n$, then $V = U \oplus W$.

(d) $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

**Why.**

1. (a), (b): an independent list extends to a basis, and every basis has length $n$.
2. (c): a basis of $U$ followed by a basis of $W$ is independent (a relation gives $u = -w \in U \cap W = \{0\}$) and has length $n$, so it is a basis by (a).
3. (d): extend a basis $x_1, \dots, x_k$ of $U \cap W$ to a basis $x_i, u_j$ of $U$ and, separately, to a basis $x_i, w_l$ of $W$. Together the $x_i, u_j, w_l$ span $U + W$; it remains to show they are independent.
4. If $x + u + w = 0$ ($x, u, w$ combinations of the $x_i$, $u_j$, $w_l$), then $w = -x - u \in U \cap W$, so $w$ is a combination of the $x_i$. The basis $x_i, w_l$ of $W$ is independent, so $w = 0$. Then $x + u = 0$, and independence of $x_i, u_j$ gives $x = u = 0$.

(T4) **Products of operators.** If $S, T$ are injective, so is $ST$. So if a product sends some $v \neq 0$ to 0, some factor is not injective.

**Why.** $STv = 0 \Rightarrow Tv = 0 \Rightarrow v = 0$.

(T5) **Polynomials in $T$ commute.** $p(T)q(T) = q(T)p(T)$. Hence $\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are $T$-invariant.

**Why.** Both products equal $(pq)(T)$. If $p(T)v = 0$, then $p(T)(Tv) = T\,p(T)v = 0$. And $T\big(p(T)u\big) = p(T)(Tu) \in \operatorname{range} p(T)$.

(T6) **Fundamental theorem of algebra.** A nonconstant $p \in \mathcal{P}(\mathbb{C})$ factors as $c(z-\lambda_1)\cdots(z-\lambda_m)$ (cited). Over any field, $p(\lambda)=0 \iff (z-\lambda) \mid p$.

**Why.** Divide $p$ by $z-\lambda$: the remainder is a constant, and evaluating at $\lambda$ shows it equals $p(\lambda)$.

(T7) **Eigenvalues.**

(a) $\lambda$ is an eigenvalue $\iff$ $T-\lambda I$ is not injective.

(b) If $Tv=\lambda v$, then $p(T)v=p(\lambda)v$.

(c) Eigenvectors for distinct eigenvalues are independent (Prop. 54).

(d) Over $\mathbb{C}$ with $V \neq 0$, every operator has an eigenvalue.

**Why.**

1. (a) is the definition: $Tv=\lambda v$ with $v\neq 0$ means $(T-\lambda I)v=0$ with $v\neq 0$.
2. (b): $T^k v=\lambda^k v$ by induction; add up the terms of $p$.
3. (d): take $v\neq 0$. The $n+1$ vectors $v, Tv, \ldots, T^n v$ are dependent, so $p(T)v=0$ for some $p\neq 0$; $p$ is not constant since $v\neq 0$.
4. Factor $p=c(z-\lambda_1)\cdots(z-\lambda_m)$ (T6). The product $c(T-\lambda_1 I)\cdots(T-\lambda_m I)$ sends $v\neq 0$ to 0, so some $T-\lambda_j I$ is not injective (T4).

(T8) **Block diagonal matrices.** If $V=U\oplus W$ with both $T$-invariant, then in a basis of $U$ followed by one of $W$, $\mathcal{M}(T)=\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$, and $\operatorname{null} T = \operatorname{null} T|_U \oplus \operatorname{null} T|_W$.

**Why.** $Tu \in U$ is a combination of the $U$-basis only, so its column is 0 in the $W$-rows (same for $W$). If $T(u+w)=0$, then $Tu=-Tw \in U\cap W=\{0\}$.

(T9) **Change of basis.** If column $j$ of $S$ holds the $B$-coordinates of the $j$-th vector of $B'$, then $\mathcal{M}_{B'}(T)=S^{-1}\mathcal{M}_B(T)S$. So similar matrices are exactly the matrices of one operator in two bases.

**Why.** $S$ converts $B'$-coordinates to $B$-coordinates: apply $S$, then $\mathcal{M}_B(T)$, then convert back with $S^{-1}$. Every invertible $S$ arises this way, since its columns form a basis.

(T10) **Quotient operator.** If $U$ is $T$-invariant, then $(T/U)(v+U)=Tv+U$ is a well-defined operator on $V/U$, $\dim V/U=\dim V-\dim U$, and $p(T/U)(v+U)=p(T)v+U$.

**Why.**

1. Well defined: if $v+U=v'+U$, then $v-v'\in U$, so $Tv-Tv'\in U$ ($U$ is invariant).
2. Dimension: $v\mapsto v+U$ maps $V$ onto $V/U$ with null space $U$; apply (T1).
3. $(T/U)^k(v+U)=T^k v+U$ by induction; add up the terms of $p$.

# §2 This week

## §2.1 Upper-triangular matrices (Ch. 12.1)

**The question.** For $A=\left[\begin{smallmatrix} 1 & 1 \\ -1 & 3 \end{smallmatrix}\right]$ over $\mathbb{C}$, every eigenvector is a multiple of $(1,1)$, so there is no eigenbasis. Is there a basis in which the matrix of $A$ is upper-triangular, and can the eigenvalue then be read off without solving an equation?

**Definition** (Def. 33)**.** A square matrix $A=(a_{ij})$ is **upper-triangular** if $a_{ij}=0$ whenever $i>j$. Its **diagonal** is $a_{11},\ldots,a_{nn}$. It is **diagonal** if every entry off the diagonal is 0.

**Example 53.** The matrix

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \\ 0 & 0 & 0 & 10 \end{pmatrix}$$

is upper-triangular with diagonal $1, 5, 8, 10$.

**Proposition** (Prop. 59 / LADR 5.39)

Let $v_1, \dots, v_n$ be a basis of $V$. The following are equivalent:

(i) the matrix of $T$ in $v_1, \dots, v_n$ is upper-triangular;

(ii) $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$ for each $j$;

(iii) $\operatorname{span}(v_1, \dots, v_j)$ is invariant under $T$ for each $j$.

**The idea**

Column $j$ of the matrix lists the coordinates of $Tv_j$: $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$. "Upper-triangular" says the entries of column $j$ below row $j$ are 0, which is the statement that $Tv_j$ uses only $v_1, \dots, v_j$. So the proposition only translates a statement about zeros in a matrix into a statement about where basis vectors are sent.

For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: $Ne_1 = 0$ and $Ne_2 = e_1$. Each basis vector is sent to a combination of itself and earlier ones, never to a later one.

*Proof.* 1. (i)⇒(ii): $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$, and $a_{ij} = 0$ for $i > j$, so $Tv_j = a_{1j}v_1 + \cdots + a_{jj}v_j$.

2. (ii)⇒(iii): for $i \le j$, $Tv_i \in \operatorname{span}(v_1, \dots, v_i) \subseteq \operatorname{span}(v_1, \dots, v_j)$. So $T$ sends each of $v_1, \dots, v_j$, and hence their span, into $\operatorname{span}(v_1, \dots, v_j)$.

3. (iii)⇒(i): $Tv_j \in \operatorname{span}(v_1, \dots, v_j)$, so in $Tv_j = a_{1j}v_1 + \cdots + a_{nj}v_n$ the coefficients $a_{j+1,j}, \dots, a_{nj}$ are 0 (coordinates in a basis are unique).

□

**Proposition** (Prop. 60 / LADR 5.41)

If $T$ has an upper-triangular matrix in some basis, the eigenvalues of $T$ are exactly the diagonal entries of that matrix.

**The idea**

$\lambda$ is an eigenvalue exactly when $T - \lambda I$ is not injective (T7). The matrix of $T - \lambda I$ is again upper-triangular, with diagonal entries $\lambda_k - \lambda$. So the question is: when is an upper-triangular matrix injective?

Try $\left(\begin{smallmatrix} 2 & 5 \\ 0 & 3 \end{smallmatrix}\right)\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = 0$. The bottom row says $3y = 0$, so $y = 0$; then the top row says $2x = 0$, so $x = 0$. Each row, read from the bottom up, forces one more coordinate to be 0, as long as the diagonal entry in that row is not 0. So a nonzero diagonal gives an injective matrix.

If instead some diagonal entry is 0, say in position $j$, then $v_j$ is sent into $\operatorname{span}(v_1, \dots, v_{j-1})$. So $j$ independent vectors $v_1, \dots, v_j$ are all sent into a space of dimension $j - 1$, and that cannot be injective.

*Proof.* Let $\lambda_1, \dots, \lambda_n$ be the diagonal entries, in the basis $v_1, \dots, v_n$. Then $\mathcal{M}(T - \lambda I)$ is upper-triangular with diagonal entries $\lambda_k - \lambda$.

1. *If $\lambda \neq \lambda_k$ for all $k$, then $T - \lambda I$ is injective.* Suppose $(T - \lambda I)v = 0$ with $v = c_1v_1 + \cdots + c_nv_n$. Only $(T - \lambda I)v_n$ has a $v_n$-component, and it is $(\lambda_n - \lambda)v_n$. So the $v_n$-coefficient of $(T - \lambda I)v$ is $(\lambda_n - \lambda)c_n$, which must be 0; hence $c_n = 0$.

2. Downward induction: suppose $c_n = \cdots = c_{k+1} = 0$, so $v = c_1v_1 + \cdots + c_kv_k$. For $i < k$, $(T - \lambda I)v_i \in \operatorname{span}(v_1, \dots, v_i)$ has no $v_k$-component, and $(T - \lambda I)v_k$ has the $v_k$-component $(\lambda_k - \lambda)v_k$. So the $v_k$-coefficient of $(T - \lambda I)v = 0$ is $(\lambda_k - \lambda)c_k$, and $c_k = 0$. Hence every $c_k = 0$, and $\lambda$ is not an eigenvalue.

3. *If $\lambda = \lambda_j$, then $T - \lambda I$ is not injective.* For $i < j$, $(T - \lambda I)v_i \in \operatorname{span}(v_1, \ldots, v_i)$ (Prop. 59). And $(T - \lambda I)v_j \in \operatorname{span}(v_1, \ldots, v_{j-1})$, because its $v_j$-coefficient is $\lambda_j - \lambda = 0$.
4. So $T - \lambda I$ maps the $j$-dimensional space $\operatorname{span}(v_1, \ldots, v_j)$ into the $(j-1)$-dimensional space $\operatorname{span}(v_1, \ldots, v_{j-1})$. By (T1) this restriction has a nonzero null space. So $\lambda$ is an eigenvalue.

□

**Example 54.** Differentiation $D$ on $\mathcal{P}_3(\mathbb{R})$ in the basis $1, x, x^2, x^3$ has matrix

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

since $Dx^k = kx^{k-1}$. It is upper-triangular with diagonal $0, 0, 0, 0$, so the only eigenvalue is 0. Each basis vector is sent to a multiple of the previous one: $x^3 \mapsto 3x^2 \mapsto 6x \mapsto 6 \mapsto 0$.

$$\underset{\text{degree 3}}{x^3} \xrightarrow{D} \underset{\text{degree 2}}{3x^2} \xrightarrow{D} \underset{\text{degree 1}}{6x} \xrightarrow{D} \underset{\text{degree 0}}{6} \xrightarrow{D} 0$$

Each application of $D$ lowers the degree by one, so $D^4 = 0$ on $\mathcal{P}_3(\mathbb{R})$ and 0 is the only eigenvalue. Read backward, the chain is the basis $1, x, x^2, x^3$ up to scaling, which is why the matrix has its nonzero entries just above the diagonal.

**Theorem** (Prop. 61 / LADR 5.47)

If $V$ is a finite-dimensional complex vector space, every $T \in \mathcal{L}(V)$ has an upper-triangular matrix in some basis.

**The idea**

By Prop. 59 we need a basis in which each $Tv_k$ lies in the span of $v_k$ and the earlier vectors. Induction suggests: find a smaller invariant subspace $U$, take such a basis of $U$ (induction), and then add more vectors. The added vectors $v$ must satisfy $Tv \in U + \operatorname{span}(v)$.

Which $U$ makes that automatic? Write $Tv = (T - \lambda I)v + \lambda v$. The second term is a multiple of $v$. The first term lies in $\operatorname{range}(T - \lambda I)$, whatever $v$ is. So take $U = \operatorname{range}(T - \lambda I)$: then *every* vector $v$ satisfies $Tv \in U + \operatorname{span}(v)$.

For induction $U$ has to be smaller than $V$, i.e. $T - \lambda I$ must not be surjective, which for operators is the same as not injective (T2). That is exactly the condition that $\lambda$ is an eigenvalue. Over $\mathbb{C}$ an eigenvalue exists (T7); this is the only place $\mathbb{C}$ is used.

*Proof.* Induction on $n = \dim V$; for $n = 1$ every matrix is upper-triangular.

1. Take an eigenvalue $\lambda$ of $T$ (T7) and let $U = \operatorname{range}(T - \lambda I)$. $U$ is invariant under $T$ (T5). $T - \lambda I$ is not injective, so it is not surjective (T2), so $\dim U < n$.
2. By induction there is a basis $u_1, \ldots, u_m$ of $U$ in which $T|_U$ is upper-triangular: $Tu_i \in \operatorname{span}(u_1, \ldots, u_i)$.
3. Extend to a basis $u_1, \ldots, u_m, v_1, \ldots, v_k$ of $V$. Then $Tv_j = (T - \lambda I)v_j + \lambda v_j$, with $(T - \lambda I)v_j \in U = \operatorname{span}(u_1, \ldots, u_m)$. So $Tv_j \in \operatorname{span}(u_1, \ldots, u_m, v_1, \ldots, v_j)$.
4. By Prop. 59 (ii)⇒(i), the matrix of $T$ in this basis is upper-triangular.

□

Over $\mathbb{R}$ this can fail. In an upper-triangular basis, $Tv_1 \in \operatorname{span}(v_1)$, so $v_1$ is an eigenvector. The rotation $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ of $\mathbb{R}^2$ has no eigenvector, so it has no upper-triangular matrix over $\mathbb{R}$. Which real operators do have one is answered by LADR 5.44 in Section 2.4.

**The question answered.** Yes. $A(1, 1) = (2, 2)$ and $Ae_2 = (1, 3) = (1, 1) + 2e_2$, so in the basis $(1, 1), e_2$ the matrix is $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$, and the only eigenvalue 2 is on the diagonal (Prop. 60).

**Shortcuts**

- **Triangular matrix: the eigenvalues are the diagonal entries.** Reason: Prop. 60. Condition: the matrix is triangular (upper or lower; a lower-triangular matrix is upper-triangular in the reversed basis). Instance:
$$\begin{pmatrix} 2 & 5 & 1 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{pmatrix} \text{ has the eigenvalues } 2, 3.$$
- **An upper-triangular basis starts with an eigenvector.** Reason: $Tv_1 \in \operatorname{span}(v_1)$ (Prop. 59). Instance: in the question, $v_1 = (1, 1)$ is the eigenvector.

**Traps**

- The eigenvalues are the diagonal entries only for a *triangular* matrix: $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has diagonal $0, 0$ but eigenvalues $\pm 1$.
- Over $\mathbb{R}$ an operator may have no upper-triangular matrix at all: the rotation by $90°$.

## §2.2 Diagonalizable operators (Ch. 12.2)

**The question.** When is there a basis in which $T$ only rescales each basis vector?

- $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$: $A(1,1) = 3(1,1)$ and $A(1,-1) = (1,-1)$. In the basis $(1,1), (1,-1)$ the matrix is $\left(\begin{smallmatrix} 3 & 0 \\ 0 & 1 \end{smallmatrix}\right)$.
- $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: every eigenvector is a multiple of $e_1$, so there are not enough eigenvectors for a basis.

**Definition** (Def. 34 / LADR 5.50)**.** $T$ is **diagonalizable** if its matrix in some basis of $V$ is diagonal. A basis of $V$ consisting of eigenvectors of $T$ is an **eigenbasis**. The **geometric multiplicity** of $\lambda$ is $\dim E(\lambda, T) = \dim \operatorname{null}(T - \lambda I)$; it is 0 when $\lambda$ is not an eigenvalue.

**Theorem** (Prop. 62 / LADR 5.55)

Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of $T$. The following are equivalent:

(i) $T$ is diagonalizable;

(ii) $V$ has a basis of eigenvectors of $T$;

(iii) $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$;

(iv) $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

**The idea**

(i) and (ii) say the same thing: column $j$ of the matrix is the coordinates of $Tv_j$, and that column is 0 except for the entry $\lambda$ in row $j$ exactly when $Tv_j = \lambda v_j$.

The content is in (iii) and (iv). The eigenspaces for different eigenvalues never overlap: their sum is always direct, because eigenvectors for distinct eigenvalues are independent (T7). So eigenspaces never give *too many* vectors. The only way to fail is to give *too few*, and that is a question of counting dimensions. That is why the single number in (iv) decides diagonalizability.

For $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$: the only eigenvalue is 0 and $\dim E(0, N) = 1 < 2$.

*Proof.* 1. (i) $\Longleftrightarrow$ (ii): the matrix in $v_1, \ldots, v_n$ is diagonal with entries $\mu_1, \ldots, \mu_n$ exactly when $Tv_j = \mu_j v_j$ for every $j$, i.e. when every $v_j$ is an eigenvector.

2. *The sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is always direct.* Suppose $u_1 + \cdots + u_m = 0$ with $u_i \in E(\lambda_i, T)$. The nonzero $u_i$ are eigenvectors for distinct eigenvalues, so they are independent (T7); a sum of independent vectors is not 0. So there are no nonzero $u_i$.

3. (ii)⇒(iii): each vector of an eigenbasis lies in some $E(\lambda_i, T)$, so the sum of the eigenspaces contains a basis of $V$ and equals $V$. It is direct by step 2.
4. (iii)⇒(iv): the dimension of a direct sum is the sum of the dimensions.
5. (iv)⇒(iii): by step 2 the sum of the eigenspaces is direct, so its dimension is $\sum \dim E(\lambda_i, T) = \dim V$. A subspace of full dimension is $V$ (T3).
6. (iii)⇒(ii): choose a basis of each $E(\lambda_i, T)$. Because the sum is direct, together they form a basis of $V$, and every vector in it is an eigenvector.

□

**Example 55: an operator that is not diagonalizable.** For every $n \geq 2$ there is an operator on $V$ that is not diagonalizable (and for $n \leq 1$ every operator is diagonal in every basis). Example: let $Tv_1 = 0$ and $Tv_k = v_{k-1}$ for $k \geq 2$, with $n \geq 2$. The matrix is upper-triangular with zero diagonal, so the only eigenvalue is 0 (Prop. 60). $\operatorname{null} T = \operatorname{span}(v_1)$: if $T(\sum c_k v_k) = \sum_{k \geq 2} c_k v_{k-1} = 0$, then $c_2 = \cdots = c_n = 0$. So $\sum_\lambda \dim E(\lambda, T) = 1 \neq n$, and $T$ is not diagonalizable by (iv).

**Proposition** (Prop. 63 / LADR 5.58)

If $T$ has $n = \dim V$ distinct eigenvalues, then $T$ is diagonalizable.

*Proof.* 1. Choose an eigenvector $v_i$ for each of the $n$ eigenvalues $\lambda_i$.

2. They are independent (T7), and there are $n$ of them, so they form a basis (T3).
3. Every vector of this basis is an eigenvector, so it is an eigenbasis, and $T$ is diagonalizable (Prop. 62).

□

The condition is sufficient, not necessary: $I$ has one eigenvalue and is diagonal.

**Example 56: diagonalizability depends on the field.** $\left[\begin{smallmatrix} 1 & 2 \\ 2 & 3 \end{smallmatrix}\right]$ has eigenvalues $2 \pm \sqrt{5}$ (the roots of $x^2 - 4x - 1$), so it is diagonalizable over $\mathbb{R}$; over $\mathbb{Q}$ it has no eigenvalues. $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ has eigenvalues $\pm i$, so it is diagonalizable over $\mathbb{C}$ but not over $\mathbb{R}$.

**Theorem** (Gershgorin disk theorem, LADR 5D)

Let $A = (a_{jk})$ be an $n \times n$ complex matrix. Every eigenvalue of $A$ lies in at least one of the disks

$$D_j = \left\{ z \in \mathbb{C} : |z - a_{jj}| \leq \sum_{k \neq j} |a_{jk}| \right\}, \qquad j = 1, \ldots, n.$$

**The idea**

Look at the largest coordinate of an eigenvector. In that row the equation $Av = \lambda v$ says that $(\lambda - a_{jj})v_j$ is a combination of the other coordinates, each at most as large as $v_j$, with the off-diagonal entries of row $j$ as coefficients.

*Proof.* 1. Let $Av = \lambda v$ with $v \neq 0$, and choose $j$ with $|v_j|$ largest; then $|v_j| > 0$.

2. Row $j$ of $Av = \lambda v$ is $\sum_k a_{jk} v_k = \lambda v_j$, that is,

$$(\lambda - a_{jj}) v_j = \sum_{k \neq j} a_{jk} v_k.$$

3. Take absolute values (triangle inequality, Week 6) and divide by $|v_j|$; since every $|v_k|/|v_j| \leq 1$,

$$|\lambda - a_{jj}| \leq \sum_{k \neq j} |a_{jk}| \frac{|v_k|}{|v_j|} \leq \sum_{k \neq j} |a_{jk}|.$$

□

Instance: for $\left[\begin{smallmatrix}10 & 1\\ 2 & -5\end{smallmatrix}\right]$ the disks are $|z-10|\le 1$ and $|z+5|\le 2$. Neither contains 0, so 0 is not an eigenvalue and the matrix is invertible. (Its eigenvalues are $\frac{5\pm\sqrt{233}}{2}\approx 10.13$, $-5.13$, one in each disk.)

### Deciding diagonalizability and diagonalizing a matrix $A$.

1. Find the eigenvalues $\lambda_i$ (for a triangular matrix, the diagonal entries).
2. For each, $\dim E(\lambda_i, A) = n - \operatorname{rank}(A-\lambda_i I)$ (T1). $A$ is diagonalizable exactly when these add up to $n$ (Prop. 62).
3. If so, find a basis of each $\operatorname{null}(A-\lambda_i I)$ and put all these vectors as the columns of $S$. Then $S^{-1}AS = D$ is diagonal, with the eigenvalues in the same order as the columns (T9).

**Diagonalizing $\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$.** Let $A=\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$. Then $A-3I=\left(\begin{smallmatrix}-1 & 1\\ 1 & -1\end{smallmatrix}\right)$ has null space $\operatorname{span}((1,1))$; $A-I=\left(\begin{smallmatrix}1 & 1\\ 1 & 1\end{smallmatrix}\right)$ has null space $\operatorname{span}((1,-1))$. Dimensions $1+1=2$. With $S=\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$: $S^{-1}AS=\left(\begin{smallmatrix}3 & 0\\ 0 & 1\end{smallmatrix}\right)$, and $A^k = S\left(\begin{smallmatrix}3^k & 0\\ 0 & 1\end{smallmatrix}\right)S^{-1}$.

**The question answered.** There is such a basis exactly when $V$ has a basis of eigenvectors, that is, when the eigenspaces have dimensions adding up to $n$ (Prop. 62). For $A$ it is $(1,1),(1,-1)$; for $N$ there is none, since $\dim E(0,N)=1<2$.

### Shortcuts

- **$n$ distinct eigenvalues: diagonalizable.** Reason: Prop. 63. Instance: $\left[\begin{smallmatrix}1 & 4\\ 0 & 2\end{smallmatrix}\right]$ has the eigenvalues $1\neq 2$, so it is diagonalizable.
- **One eigenvalue only: diagonalizable exactly when $T=\lambda I$.** Reason: in an eigenbasis the matrix is diagonal with every entry $\lambda$, which is $\lambda I$, and $\lambda I$ has the matrix $\lambda I$ in every basis. Instance: $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ is not diagonalizable.
- **Dimension of an eigenspace by rank.** $\dim E(\lambda, A) = n - \operatorname{rank}(A-\lambda I)$. Reason: (T1). Instance:

$$A=\begin{pmatrix}1 & 1 & 0\\ 0 & 1 & 0\\ 0 & 0 & 2\end{pmatrix},\quad \operatorname{rank}(A-I)=2,\quad \text{so } \dim E(1,A)=1.$$

- **If $T$ is diagonalizable, so is every $p(T)$.** Reason: an eigenbasis of $T$ is one of $p(T)$, since $p(T)v=p(\lambda)v$ (T7(b)). Instance: $A=\left[\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right]$ is diagonalizable, so $A^2-A$ is, with eigenvalues $3^2-3=6$ and $1^2-1=0$.
- **Gershgorin for invertibility.** If no Gershgorin disk contains 0, then $A$ is invertible. Reason: the Gershgorin disk theorem and (T2). Instance: the matrix after the theorem.

### Traps

- Repeated diagonal entries decide nothing by themselves: $I$ and $\left[\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right]$ both have diagonal $1,1$; the first is diagonalizable, the second is not.
- Upper-triangular is not the same as diagonalizable. $\left[\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right]$ is upper-triangular, but it has only the eigenvalue 0 and is not 0, so it is not diagonalizable.
- Diagonalizability depends on the field (Ex. 56).
- $T^2$ diagonalizable does not make $T$ diagonalizable: $N=\left[\begin{smallmatrix}0 & 1\\ 0 & 0\end{smallmatrix}\right]$ has $N^2=0$, which is diagonal, and $N$ is not diagonalizable.

## §2.3 Iteration and eigenmodes (Ch. 12.3)

**The question.** For the three-state Markov chain $P$ below, started at $p_0=(1,0,0)$, what is $P^k p_0$ exactly for every $k$, and how fast does it approach $(\frac{1}{4},\frac{1}{2},\frac{1}{4})$?

**Proposition** (Prop. 64: iteration in an eigenbasis)
If $v_1, \dots, v_n$ is an eigenbasis with $Tv_j = \lambda_j v_j$ and $v = a_1 v_1 + \cdots + a_n v_n$, then $T^k v = a_1 \lambda_1^k v_1 + \cdots + a_n \lambda_n^k v_n$ for every $k \geq 0$.

*Proof.* 1. $T^k v_j = \lambda_j^k v_j$ by induction on $k$: $T^0 v_j = v_j$, and $T^{k+1} v_j = T(\lambda_j^k v_j) = \lambda_j^k T v_j = \lambda_j^{k+1} v_j$.

2. $T^k$ is linear, so $T^k v = \sum_j a_j T^k v_j = \sum_j a_j \lambda_j^k v_j$.

□

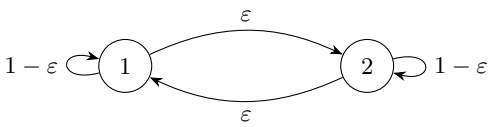
The notes call each $a_j v_j$ an **eigenmode**. Under repeated application of $T$, a mode with $|\lambda_j| < 1$ shrinks to 0, one with $|\lambda_j| = 1$ keeps its size, and one with $|\lambda_j| > 1$ grows.

**Markov chain.** Take the stochastic matrix $P$ and the three vectors

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{pmatrix}, \qquad \pi = \begin{pmatrix} 1/4 \\ 1/2 \\ 1/4 \end{pmatrix}, \quad u = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

1. Multiplying: $P\pi = \pi$, $Pu = \frac{1}{2}u$, $Pw = 0$. So $\pi, u, w$ is an eigenbasis with eigenvalues $1, \frac{1}{2}, 0$.
2. The start $p_0 = (1, 0, 0)$ is $\pi - \frac{1}{2}u + \frac{1}{4}w$ (check the first coordinate: $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$).
3. By Prop. 64, for $k \geq 1$:
$$P^k p_0 = \pi - \frac{1}{2} \cdot \frac{1}{2^k} u + 0 = \pi - 2^{-(k+1)} u.$$
4. So $P^k p_0 \to \pi$, and the distance to $\pi$ halves at every step. The eigenvalue 1 gives the limit; the next largest eigenvalue, $\frac{1}{2}$, gives the speed.

- **Spectral gap.** $P_\varepsilon = \left(\begin{smallmatrix} 1-\varepsilon & \varepsilon \\ \varepsilon & 1-\varepsilon \end{smallmatrix}\right)$, $0 < \varepsilon < \frac{1}{2}$, has eigenvectors $(1, 1)$ and $(1, -1)$ with eigenvalues 1 and $1 - 2\varepsilon$. So $P_\varepsilon^k(1, -1) = (1 - 2\varepsilon)^k(1, -1)$. When $\varepsilon$ is small (moving between the two states is rare), $1 - 2\varepsilon$ is close to 1 and the starting state is forgotten slowly. The number $1 - (1 - 2\varepsilon) = 2\varepsilon$ is the **spectral gap**. A narrow route between two large groups of states, like the small probability $\varepsilon$ here, is called a **bottleneck**; a bottleneck gives a small spectral gap and slow convergence.




The chain $P_\varepsilon$: each step stays put with probability $1 - \varepsilon$. The difference between the two probabilities is multiplied by $1 - 2\varepsilon$ at each step, so for small $\varepsilon$ it takes about $1/(2\varepsilon)$ steps to shrink noticeably.

- **PageRank** (Week 6). There $P$ is the column-stochastic link matrix, $\alpha \in (0, 1)$ the damping factor, and $G$ the Google matrix, with $\mathbb{R}^n = \operatorname{span}(r) \oplus H$, $Gr = r$, and $Gx = \alpha P x$ for $x \in H$. If $x \in H$ is an eigenvector of $P$ with eigenvalue $\lambda$, then $Gx = \alpha\lambda x$. Every eigenvalue of $P$ has modulus at most 1, so these eigenvalues of $G$ have modulus at most $\alpha$: in an eigenbasis, every mode other than $r$ is multiplied by a factor of modulus at most $\alpha$ per step. (The contraction estimate of Week 6 gave the same convergence without assuming that $P$ is diagonalizable.)

The sequences $(a_0, a_1, a_2, \dots)$ with $a_{k+1} = a_k + a_{k-1}$ form a subspace $R$, and each one is determined by $(a_0, a_1)$. So $J(a_0, a_1, \dots) = (a_0, a_1)$ is an isomorphism $R \to \mathbb{R}^2$. Under $J$ the shift $S(a_0, a_1, a_2, \dots) = (a_1, a_2, \dots)$ becomes $T(x, y) = (y, x + y)$, so a question about an infinite sequence becomes a question about powers of one operator on $\mathbb{R}^2$.

**Fibonacci numbers, LADR 5D Exercise 21.** $F_0 = 0$, $F_1 = 1$, $F_{k+1} = F_k + F_{k-1}$. Let $T(x, y) = (y, x + y)$, so $T(F_{k-1}, F_k) = (F_k, F_{k+1})$ and $T^k(0, 1) = (F_k, F_{k+1})$.

1. $T(1, \lambda) = (\lambda, 1 + \lambda)$ equals $\lambda(1, \lambda)$ exactly when $\lambda^2 = \lambda + 1$: $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Two distinct eigenvalues, so $(1, \varphi), (1, \psi)$ is an eigenbasis (Prop. 63).

2. $(0,1) = \frac{1}{\sqrt{5}}\big((1,\varphi) - (1,\psi)\big)$, since $\varphi - \psi = \sqrt{5}$.

3. By Prop. 64, $T^k(0,1) = \frac{1}{\sqrt{5}}\big(\varphi^k(1,\varphi) - \psi^k(1,\psi)\big)$. First coordinate: $F_k = \frac{\varphi^k - \psi^k}{\sqrt{5}}$.

**The question answered.** $P^k p_0 = \pi - 2^{-(k+1)}u$ for $k \geq 1$ (the Markov example), so the distance to $\pi$ is halved at every step.

**Shortcuts**

- **Powers.** Write $v$ in an eigenbasis and multiply each coefficient by $\lambda_j^k$, or use $A^k = SD^kS^{-1}$. Reason: Prop. 64 and (T9). Instance: for $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$,

$$A^k = \frac{1}{2}\begin{pmatrix} 3^k + 1 & 3^k - 1 \\ 3^k - 1 & 3^k + 1 \end{pmatrix}.$$

- **Speed of convergence: the second-largest $|\lambda|$.** Reason: Prop. 64; every mode with $|\lambda_j| < 1$ is multiplied by $\lambda_j$ at each step. Instance: $P_\varepsilon$, where the factor is $1 - 2\varepsilon$.

**Traps**

- An eigenvalue of modulus 1 other than 1 prevents convergence: $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has the eigenvalue $-1$ with eigenvector $(1,-1)$, and $P^k e_1$ alternates between $e_1$ and $e_2$.

## §2.4 Minimal polynomial (Ch. 13)

**The question.** For $A = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $A^2 = 4A - 3I$. Can every power of $A$ be written as a combination of $I$ and $A$? Which polynomials $p$ have $p(A) = 0$, and what do they tell us about the eigenvalues of $A$ and whether $A$ is diagonalizable?

**Definition** (Def. 35, 36 / LADR 5.14, 5.22, 5.24)**.** For $p(x) = a_m x^m + \cdots + a_1 x + a_0$, $p(T) = a_m T^m + \cdots + a_1 T + a_0 I$. Let $k$ be the smallest index with $T^k \in \operatorname{span}(I, T, \ldots, T^{k-1})$, and write $T^k = a_{k-1}T^{k-1} + \cdots + a_0 I$. The **minimal polynomial** of $T$ is $m_T(x) = x^k - a_{k-1}x^{k-1} - \cdots - a_0$. (The notes call it $p$.)

- $k$ exists: $\mathcal{L}(V)$ has dimension $n^2$ (operators are $n \times n$ matrices), so the $n^2 + 1$ operators $I, T, \ldots, T^{n^2}$ are dependent. Solving a dependence for the highest power with a nonzero coefficient puts that power in the span of the lower ones.
- $I, T, \ldots, T^{k-1}$ are independent, by the same argument: a dependence among them would put some $T^j$ with $j < k$ in the span of lower powers. So the $a_i$ are unique.
- $m_T$ is monic and $m_T(T) = 0$. No nonzero polynomial of degree $< k$ kills $T$, since it would be a dependence among $I, \ldots, T^{k-1}$. So $m_T$ is the unique monic polynomial of least degree that kills $T$ (two of them would differ by a nonzero polynomial of lower degree that kills $T$, and dividing it by its leading coefficient would give a monic one).

**Example 57.** $m_I = x - 1$. For $T = \left[\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right]$, $I, T$ are independent ($T$ is not a multiple of $I$) and

$$T^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = 4T - 3I,$$

so $m_T = x^2 - 4x + 3 = (x-1)(x-3)$.

**Proposition** (Prop. 65 / LADR 5.29)
For $q \in \mathcal{P}(\mathbb{F})$: $q(T) = 0 \iff m_T \mid q$.

**The idea**

$m_T$ has the smallest degree among nonzero polynomials that kill $T$. Polynomials can be divided with remainder, and the remainder has smaller degree than the divisor.

Divide $q$ by $m_T$. The remainder kills $T$ too, and its degree is too small for it to be nonzero.

*Proof.* 1. If $q = m_T r$, then $q(T) = m_T(T)r(T) = 0$ (T5).

2. If $q(T) = 0$, divide: $q = m_T r + s$ with $s = 0$ or $\deg s < \deg m_T$. Then $s(T) = q(T) - m_T(T)r(T) = 0$.

3. A nonzero $s$ of degree $< \deg m_T$ cannot kill $T$. So $s = 0$ and $m_T \mid q$.

□

**Proposition** (Prop. 66 / LADR 5.27)
The eigenvalues of $T$ are exactly the roots of $m_T$ in $\mathbb{F}$. If $m_T = (x - \lambda)s$, then $s(T) \neq 0$, and every nonzero vector $s(T)w$ is an eigenvector for $\lambda$.

**The idea**

On an eigenvector, a polynomial in $T$ acts as the number $p(\lambda)$ (T7). Apply this to $m_T$, which is 0 as an operator.

For the converse we need a nonzero vector sent to 0 by $T - \lambda I$. Factor $m_T = (x - \lambda)s$. Then $(T - \lambda I)s(T) = m_T(T) = 0$, so $T - \lambda I$ kills every vector of the form $s(T)w$. Such a vector can be chosen nonzero because $s$ has smaller degree than $m_T$, so $s(T) \neq 0$.

*Proof.* 1. If $Tv = \lambda v$ with $v \neq 0$, then $0 = m_T(T)v = m_T(\lambda)v$, so $m_T(\lambda) = 0$.

2. If $m_T(\lambda) = 0$, write $m_T = (x - \lambda)s$ (T6). Since $\deg s < \deg m_T$, $s(T) \neq 0$; pick $w$ with $s(T)w \neq 0$.

3. $(T - \lambda I)\big(s(T)w\big) = m_T(T)w = 0$, so $s(T)w$ is an eigenvector for $\lambda$.

The notes argue the second half by contradiction: if $T - \lambda I$ were invertible, multiplying $0 = s(T)(T - \lambda I)$ by $(T - \lambda I)^{-1}$ would give $s(T) = 0$. □

**Proposition** (Divisors of a product of linear factors)
If a monic polynomial $s$ divides $(x - \lambda_1) \cdots (x - \lambda_r)$, with $\lambda_i \in \mathbb{F}$ not necessarily distinct, then $s$ is the product of some of these factors.

*Proof.* 1. Induction on $r$; for $r = 0$ the product is 1 and $s = 1$. Write $st = (x - \lambda_1) \cdots (x - \lambda_r)$.

2. At $x = \lambda_r$ the right side is 0, so $s(\lambda_r) = 0$ or $t(\lambda_r) = 0$. By the factor theorem (T6), $x - \lambda_r$ divides $s$ or $t$: $s = (x - \lambda_r)s_1$ or $t = (x - \lambda_r)t_1$.

3. Cancel the factor $x - \lambda_r$ from both sides. (If $(x - \lambda)a = (x - \lambda)b$, then $(x - \lambda)(a - b) = 0$, and a nonzero $a - b$ would give a nonzero product, so $a = b$.) This leaves $s_1 t = (x - \lambda_1) \cdots (x - \lambda_{r-1})$ or $st_1 = (x - \lambda_1) \cdots (x - \lambda_{r-1})$.

4. By induction $s_1$, respectively $s$, is a product of some of $x - \lambda_1, \ldots, x - \lambda_{r-1}$. So $s$ is a product of some of $x - \lambda_1, \ldots, x - \lambda_r$.

□

**Proposition** (LADR 5.40, 5.44: upper-triangular matrices and $m_T$)

(a) If $\mathcal{M}(T)$ is upper-triangular in $v_1, \dots, v_n$ with diagonal $\lambda_1, \dots, \lambda_n$, then $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$. (b) $T$ has an upper-triangular matrix in some basis if and only if $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$ for some $\lambda_i \in \mathbb{F}$ (not necessarily distinct).

**The idea**

In (a), $T - \lambda_j I$ sends $\operatorname{span}(v_1, \dots, v_j)$ into $\operatorname{span}(v_1, \dots, v_{j-1})$, because the $v_j$-coefficient of $(T - \lambda_j I)v_j$ is $\lambda_j - \lambda_j = 0$. Applying $T - \lambda_n I$, then $T - \lambda_{n-1} I$, and so on lowers the span by one each time, and after $n$ steps everything is 0. For (b, $\Leftarrow$), copy the proof of Prop. 61, but get the eigenvalue from a root of $m_T$ (Prop. 66) instead of from $\mathbb{C}$.

*Proof.* (a)

1. Let $U_j = \operatorname{span}(v_1, \dots, v_j)$ and $U_0 = \{0\}$. For $i < j$, $(T - \lambda_j I)v_i \in U_i \subseteq U_{j-1}$ (Prop. 59), and $(T - \lambda_j I)v_j \in U_{j-1}$ since its $v_j$-coefficient is 0. So $(T - \lambda_j I)U_j \subseteq U_{j-1}$.
2. The factors commute (T5), so apply them to $v \in V = U_n$ in the order $T - \lambda_n I$, then $T - \lambda_{n-1} I$, …, then $T - \lambda_1 I$: the result lies in $U_{n-1}$, then $U_{n-2}$, …, then $U_0 = \{0\}$.

(b)

1. ($\Rightarrow$) By (a) the product $(x - \lambda_1) \cdots (x - \lambda_n)$ of the diagonal entries kills $T$, so $m_T$ divides it (Prop. 65), and $m_T$ is a product of some of its factors (the previous proposition).
2. ($\Leftarrow$) Induction on $n$; for $n = 1$ every matrix is upper-triangular. $m_T$ has degree $\geq 1$ and is a product of linear factors, so it has a root $\lambda \in \mathbb{F}$, which is an eigenvalue (Prop. 66).
3. $U = \operatorname{range}(T - \lambda I)$ is invariant (T5), and $\dim U < n$ because $T - \lambda I$ is not injective (T2). For $u \in U$, $m_T(T|_U)u = m_T(T)u = 0$, so $m_{T|_U} \mid m_T$ (Prop. 65 for $T|_U$), and $m_{T|_U}$ is again a product of linear factors (the previous proposition).
4. By induction $T|_U$ is upper-triangular in a basis $u_1, \dots, u_m$ of $U$ (empty if $U = \{0\}$). Extend it to a basis $u_1, \dots, u_m, v_1, \dots, v_k$ of $V$. Then $Tv_j = (T - \lambda I)v_j + \lambda v_j \in \operatorname{span}(u_1, \dots, u_m, v_j)$, and Prop. 59 gives an upper-triangular matrix.

□

Instance: over $\mathbb{R}$ the rotation has $m = x^2 + 1$, which is not a product of real linear factors, so it has no upper-triangular matrix over $\mathbb{R}$.

**Proposition** (Prop. 67 / LADR 5.22)

$\deg m_T \leq n$.

**The idea**

It is enough to find some polynomial of degree $\leq n$ that kills $T$, since $m_T$ divides it (Prop. 65).

For one vector $v \neq 0$, the $n + 1$ vectors $v, Tv, \dots, T^n v$ are dependent, so $q(T)v = 0$ for some $q \neq 0$ of degree $\leq n$. This kills $v$, but not necessarily all of $V$.

A first attempt is to do this for each basis vector and multiply the $q$'s. The product kills $T$, but its degree can be about $n^2$.

Induction on $\dim V$. $U = \operatorname{null} q(T)$ is invariant and contains $v$. If $U \neq V$, both $U$ and $V/U$ are smaller than $V$. By induction, a polynomial $r$ of degree $\leq \dim U$ kills $T$ on $U$, and a polynomial $p$ of degree $\leq \dim V/U$ kills $T/U$, which means $p(T)$ sends $V$ into $U$. So $r(T)p(T)$ kills $V$, and its degree is $\leq \dim U + \dim V/U = n$.

*Proof.* 1. If $\dim V = 1$, $T = cI$ and $m_T = x - c$. In general, fix $v \neq 0$ and $q \neq 0$ with $\deg q \leq n$ and $q(T)v = 0$. $U = \operatorname{null} q(T)$ is invariant (T5).

2. If $q(T) = 0$, then $m_T \mid q$, so $\deg m_T \leq n$.
3. Otherwise $0 < \dim U < n$. By induction, $r = m_{T|_U}$ has $\deg r \leq \dim U$ and $p = m_{T/U}$ has $\deg p \leq \dim V/U$ (T10).

4. For $v \in V$: $p(T)v + U = p(T/U)(v + U) = 0 + U$, so $p(T)v \in U$, so $r(T)p(T)v = 0$. Hence $m_T \mid rp$, and $\deg(rp) \le \dim U + \dim V/U = n$ (T10).

□

**Second proof (LADR 5.22), without quotient spaces.** Same plan, with $\operatorname{range} q(T)$ in place of $V/U$. Induction on $n$.

1. Fix $v \neq 0$ and let $q$ be the monic polynomial of least degree $d$ with $q(T)v = 0$; $d \ge 1$ since $v \neq 0$.
2. $v, Tv, \ldots, T^{d-1}v$ are independent (a dependence would be a smaller polynomial killing $v$), and they lie in $\operatorname{null} q(T)$, because $q(T)T^j v = T^j q(T)v = 0$. So $\dim \operatorname{null} q(T) \ge d$.
3. $W = \operatorname{range} q(T)$ is invariant (T5) and $\dim W \le n - d < n$ (T1). By induction some $s$ of degree $\le n - d$ kills $T|_W$.
4. For every $u$: $q(T)u \in W$, so $s(T)q(T)u = 0$. So $sq$ kills $T$, and $\deg(sq) \le n$.

**Example 58: every degree from 1 to $n$ occurs.**

- $m_I = x - 1$ in every dimension.
- The shift $N$ ($Ne_1 = 0$, $Ne_j = e_{j-1}$) has $m_N = x^n$, since $N^{n-1}e_n = e_1 \neq 0$ and $N^n = 0$.
- Degree $k$, $1 \le k \le n$: let $Ne_1 = 0$, $Ne_j = e_{j-1}$ for $2 \le j \le k$, $Ne_j = 0$ for $j > k$. Then $N^k = 0$ and $N^{k-1}e_k = e_1 \neq 0$, so $m_N = x^k$.

**Proposition** (Prop. 68 / LADR 5.62)

$T$ is diagonalizable if and only if $m_T = (x - \lambda_1) \cdots (x - \lambda_m)$ for some distinct $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

**The idea**

($\Rightarrow$) Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues. On an eigenvector for $\lambda_j$, the factor $T - \lambda_j I$ gives 0, and the factors commute, so that factor can act first. So $\prod_i (T - \lambda_i I)$ sends every vector of an eigenbasis to 0, and it is the zero map.

($\Leftarrow$) We must write every $v$ as a sum of eigenvectors. Since $m_T(T) = 0$, removing one factor $T - \lambda_j I$ from $m_T(T)$ leaves a map whose outputs are eigenvectors for $\lambda_j$ (or 0). It remains to choose these pieces so that they add up to $v$.

**Producing the eigenvectors.** Factor $T - \lambda_j I$ out of $m_T(T) = 0$:

$$(T - \lambda_j I) \prod_{i \neq j} (T - \lambda_i I) = m_T(T) = 0.$$

So every vector of the form $\prod_{i \neq j}(T - \lambda_i I)w$ is in $E(\lambda_j, T)$. To make the pieces add up to $v$, scale with the polynomials

$$g_j(x) = \prod_{i \neq j} \frac{x - \lambda_i}{\lambda_j - \lambda_i},$$

which are 1 at $\lambda_j$ and 0 at the other $\lambda_i$, and satisfy $\sum_j g_j = 1$ (step 2 of the proof). Applying both sides to $v$ gives $v = \sum_j g_j(T)v$, and each $g_j(T)v$ is a multiple of a vector of the form $\prod_{i \neq j}(T - \lambda_i I)w$, so it lies in $E(\lambda_j, T)$.

*Proof.* 1. ($\Rightarrow$) Let $v_1, \ldots, v_n$ be an eigenbasis, $Tv_k = \mu v_k$ with $\mu$ one of the $\lambda_i$. Letting the factor $T - \mu I$ act first, $\prod_i (T - \lambda_i I)v_k = 0$ for every $k$ (T5). So $m_T \mid \prod_i (x - \lambda_i)$ (Prop. 65). Each $\lambda_i$ is a root of $m_T$ (Prop. 66), so $m_T = \prod_i (x - \lambda_i)$.

2. ($\Leftarrow$) Define $g_j$ as above. $\sum_j g_j - 1$ has degree $\le m - 1$ and vanishes at the $m$ points $\lambda_1, \ldots, \lambda_m$. A nonzero polynomial of degree $\le m - 1$ has at most $m - 1$ roots (each root splits off a factor, T6), so $\sum_j g_j = 1$.

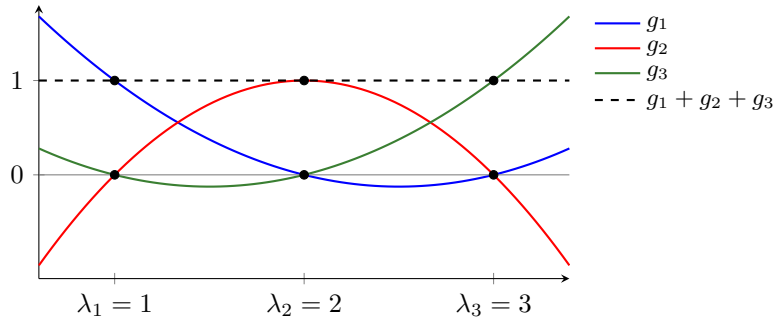
3. For $v \in V$: $v = \sum_j g_j(T)v$, and $(T - \lambda_j I)g_j(T)v = c_j\, m_T(T)v = 0$, where

$$c_j = \prod_{i \neq j} (\lambda_j - \lambda_i)^{-1}.$$

So $g_j(T)v \in E(\lambda_j, T)$.

4. So $V = E(\lambda_1, T) + \cdots + E(\lambda_m, T)$. This sum is direct (T7), so bases of the eigenspaces together form an eigenbasis.

□




$g_j$ is 1 at $\lambda_j$ and 0 at the other eigenvalues, and $g_1 + g_2 + g_3 = 1$. Applying $1 = \sum g_j$ to $T$ writes $v = \sum g_j(T)v$, and each $g_j(T)v$ lies in $E(\lambda_j, T)$ (Prop. 68).

**Second proof of (⇐), the one in the notes.** Instead of splitting $v$ into all its eigenvector pieces at once, split off the $\lambda_1$-piece and use induction for the rest. Let $g = \prod_{i \geq 2}(x - \lambda_i)$, so $m_T = (x - \lambda_1)g$ and $g(\lambda_1) \neq 0$.

1. Dividing by $x - \lambda_1$ leaves the remainder $g(\lambda_1)$ (T6): $g = (x - \lambda_1)q + g(\lambda_1)$.
2. For $v \in V$ let $u = g(T)v/g(\lambda_1)$. Then $(T - \lambda_1 I)u = m_T(T)v/g(\lambda_1) = 0$, so $u \in \text{null}(T - \lambda_1 I)$, and $v - u = -(T - \lambda_1 I)q(T)v/g(\lambda_1) \in \text{range}(T - \lambda_1 I)$.
3. So $V = \text{null}(T - \lambda_1 I) + \text{range}(T - \lambda_1 I)$. The two dimensions add up to $n$ (T1), so the intersection is 0 (last item of T3) and the sum is direct.
4. $\text{range}(T - \lambda_1 I)$ is invariant, smaller than $V$ (since $\text{null}(T - \lambda_1 I) \neq 0$: $\lambda_1$ is an eigenvalue by Prop. 66), and its minimal polynomial divides $m_T$, because $m_T(T|_U)u = m_T(T)u = 0$ for $u$ in $U = \text{range}(T - \lambda_1 I)$ (then Prop. 65 for $T|_U$). By the proposition on divisors of a product of linear factors, it again is a product of distinct linear factors. By induction $T|_U$ has an eigenbasis; add a basis of $E(\lambda_1, T)$.

**The question answered.** Every power of $A$ is a combination of $I$ and $A$: $A^k$ is the remainder of $x^k$ on division by $m_A$, evaluated at $A$ (Shortcuts, "Any polynomial"). The polynomials with $p(A) = 0$ are the multiples of $m_A = (x-1)(x-3)$ (Prop. 65). Its roots $1, 3$ are the eigenvalues (Prop. 66), and they are distinct, so $A$ is diagonalizable (Prop. 68).

### More facts about $m_T$

Throughout, write $m_T = x^d + c_{d-1}x^{d-1} + \cdots + c_1 x + c_0$.

**Proposition** (F1: invertibility, HW7 P3)
$T$ is invertible $\iff$ $c_0 \neq 0$. In that case $T^{-1} = -\frac{1}{c_0}\left(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\right)$, a polynomial in $T$.

**The idea**

$m_T(T) = 0$, and every term of $m_T(T)$ except $c_0 I$ contains a factor $T$.

Move $c_0 I$ to the other side and factor out $T$. That gives $T \cdot (\text{something}) = -c_0 I$, so $\frac{-1}{c_0}(\text{something})$ is the inverse.

*Proof.* 1. $c_0 = m_T(0)$. By Prop. 66, $c_0 = 0$ $\iff$ 0 is an eigenvalue $\iff$ $T$ is not injective (T7) $\iff$ $T$ is not invertible (T2).

2. If $c_0 \neq 0$: $m_T(T) = 0$ says $T^d + c_{d-1}T^{d-1} + \cdots + c_1 T = -c_0 I$.
3. Factor $T$ out of the left side: $T\left(T^{d-1} + c_{d-1}T^{d-2} + \cdots + c_1 I\right) = -c_0 I$. Divide by $-c_0$.

□

**Proposition** (F2: degree one, M4)
If $V \neq \{0\}$: $\deg m_T = 1 \iff T = cI$ for some $c \in \mathbb{F}$.

*Proof.* 1. ($\Rightarrow$) $m_T = x - c$ and $m_T(T) = 0$ say $T - cI = 0$.

2. ($\Leftarrow$) $x - c$ kills $cI$. No polynomial of degree 0 does: a nonzero constant $a$ gives $aI \neq 0$ since $V \neq \{0\}$. So $m_{cI} = x - c$.

□

**Proposition** (F3: restrictions and quotients, Ex. 60)
Let $U$ be $T$-invariant.

(a) $m_{T|_U} \mid m_T$ and $m_{T/U} \mid m_T$.

(b) A monic divisor of a product $(x - \lambda_1) \cdots (x - \lambda_m)$ of distinct linear factors is the product of some of those factors.

(c) So if $T$ is diagonalizable, so are $T|_U$ and $T/U$.

*Proof.* (a)

1. For $u \in U$: $m_T(T|_U)u = m_T(T)u = 0$. So $m_T$ kills $T|_U$, and $m_{T|_U} \mid m_T$ by Prop. 65 applied to $T|_U$.
2. For $v + U \in V/U$: $m_T(T/U)(v + U) = m_T(T)v + U = 0 + U$ (T10). So $m_{T/U} \mid m_T$.

(b) Let $s$ be the monic divisor, $st = \prod_{i=1}^{m}(x - \lambda_i)$.

1. Each $\lambda_i$ makes the product 0, so it is a root of $s$ or of $t$. Let $S$ be the set of $\lambda_i$ that are roots of $s$, and $R$ the remaining $\lambda_i$ (roots of $t$).
2. Distinct roots split off one factor each (T6), so $\prod_{\lambda \in S}(x - \lambda)$ divides $s$ and $\prod_{\lambda \in R}(x - \lambda)$ divides $t$. Hence $\deg s \geq |S|$ and $\deg t \geq |R|$.
3. But $\deg s + \deg t = m = |S| + |R|$, so both are equalities. A monic $s$ divisible by a monic polynomial of the same degree equals it: $s = \prod_{\lambda \in S}(x - \lambda)$.

(c) $m_T$ is a product of distinct linear factors (Prop. 68). By (a) and (b), so are $m_{T|_U}$ and $m_{T/U}$. Apply Prop. 68 again. □

**Definition** (Least common multiple)**.** For nonzero polynomials $p_1, \ldots, p_r$, a **common multiple** is a polynomial divisible by every $p_j$. The **least common multiple** $\operatorname{lcm}(p_1, \ldots, p_r)$ is the monic common multiple of least degree.

**Proposition** (The lcm divides every common multiple)
$L = \operatorname{lcm}(p_1, \ldots, p_r)$ exists, is unique, and divides every common multiple. If $p_j = \prod_\lambda (x - \lambda)^{e_{j\lambda}}$ for all $j$, then $L = \prod_\lambda (x - \lambda)^{\max_j e_{j\lambda}}$.

*Proof.* 1. The product $p_1 \cdots p_r$ is a common multiple, so a monic common multiple $L$ of least degree exists.

2. Let $q$ be any common multiple and divide: $q = Lh + s$ with $s = 0$ or $\deg s < \deg L$.
3. Each $p_j$ divides $q$ and $Lh$, hence divides $s = q - Lh$. A nonzero $s$, divided by its leading coefficient, would be a monic common multiple of smaller degree than $L$, which is impossible. So $s = 0$ and $L \mid q$. Two such $L$ divide each other and are monic, so they are equal.
4. For the formula: $\prod_\lambda (x - \lambda)^{\max_j e_{j\lambda}}$ is a common multiple, so $L$ divides it, and $L$ is a product of some of its factors (divisors of a product of linear factors). $L$ is divisible by each $p_j$, hence by $(x - \lambda)^{e_{j\lambda}}$ for every $j$; so the exponent of $x - \lambda$ in $L$ is at least $\max_j e_{j\lambda}$. Hence equality.

□

**Proposition** (F4: direct sums, block diagonal matrices)

If $V = U_1 \oplus \cdots \oplus U_r$ with every $U_i$ invariant, then $m_T = \operatorname{lcm}\big(m_{T|_{U_1}}, \ldots, m_{T|_{U_r}}\big)$. For a block diagonal matrix, $m$ is the lcm of the minimal polynomials of the blocks.

*Proof.* 1. Write $v = u_1 + \cdots + u_r$ with $u_i \in U_i$. Then $q(T)v = \sum_i q(T|_{U_i})u_i$, so $q(T) = 0 \iff q(T|_{U_i}) = 0$ for every $i$.

2. By Prop. 65, that holds $\iff$ every $m_{T|_{U_i}}$ divides $q$.

3. So $m_T$ is the monic polynomial of least degree divisible by all the $m_{T|_{U_i}}$: their lcm. The blocks of the matrix are the matrices of the $T|_{U_i}$ (T8).

□

**Proposition** (F5: similarity, transpose, dual)

Similar matrices have the same minimal polynomial. Also $m_{A^\intercal} = m_A$ and $m_{T'} = m_T$.

*Proof.* 1. $(S^{-1}AS)^k = S^{-1}A^kS$ (the inner factors $SS^{-1}$ cancel), so $p(S^{-1}AS) = S^{-1}p(A)S$. This is 0 exactly when $p(A) = 0$. So the same polynomials kill both, and the minimal polynomials agree.

2. $(A^\intercal)^k = (A^k)^\intercal$, so $p(A^\intercal) = p(A)^\intercal$, which is 0 exactly when $p(A) = 0$.

3. In dual bases, $\mathcal{M}(T') = \mathcal{M}(T)^\intercal$; apply step 2.

□

**Proposition** (F6: shifts, scalings, inverses, HW7 P4, M5)

With $d = \deg m_T$:

(a) $m_{T-\lambda I}(x) = m_T(x + \lambda)$;

(b) $m_{cT}(x) = c^d\, m_T(x/c)$ for $c \neq 0$;

(c) if $T$ is invertible, $m_{T^{-1}}(x) = \frac{1}{c_0}x^d m_T(1/x)$, i.e. reverse the list of coefficients of $m_T$ and divide by $c_0$.

**The idea**

A polynomial in the new operator is a polynomial in $T$ after a substitution: $q(T - \lambda I)$ is $r(T)$ with $r(x) = q(x - \lambda)$.

The substitution can be undone and does not change degrees. So it matches the polynomials that kill the new operator with the polynomials that kill $T$, and the one of least degree goes to the one of least degree.

*Proof.* (a)

1. For any $q$, $q(T - \lambda I) = r(T)$ with $r(x) = q(x - \lambda)$.

2. So $q$ kills $T - \lambda I \iff m_T(x) \mid q(x - \lambda)$ (Prop. 65) $\iff m_T(x + \lambda) \mid q(x)$ (replace $x$ by $x + \lambda$ throughout).

3. $m_T(x + \lambda)$ is monic, so it is $m_{T-\lambda I}$.

(b)

1. For any $q$, $q(cT) = r(T)$ with $r(x) = q(cx)$.

2. So $q$ kills $cT \iff m_T(x) \mid q(cx)$ (Prop. 65) $\iff m_T(x/c) \mid q(x)$ (replace $x$ by $x/c$ throughout; this substitution can be undone and keeps degrees).

3. $m_T(x/c)$ has leading coefficient $c^{-d}$, so $c^d m_T(x/c)$ is monic, divides the same polynomials, and is $m_{cT}$.

(c)

1. Let
$$r(x) = \frac{1}{c_0}x^d m_T(1/x) = \frac{1}{c_0}\big(c_0x^d + c_1x^{d-1} + \cdots + c_{d-1}x + 1\big).$$
It is monic of degree $d$ ($c_0 \neq 0$ by F1).

2. Substitute $T^{-1}$ into $r$ and factor out $T^{-d}$:

$$r(T^{-1}) = \frac{1}{c_0}\left(c_0 T^{-d} + c_1 T^{-(d-1)} + \cdots + I\right) = \frac{1}{c_0} T^{-d} m_T(T) = 0.$$

   So $m_{T^{-1}} \mid r$ and $\deg m_{T^{-1}} \leq d$.

3. The same argument with $T^{-1}$ in place of $T$ (its inverse is $T$) gives $d \leq \deg m_{T^{-1}}$.

4. So $m_{T^{-1}}$ divides $r$ and has the same degree; both are monic, so they are equal.

□

**Proposition** (F7: one vector at a time)

For $v \in V$, let $p_v$ be the monic polynomial of least degree with $p_v(T)v = 0$.

(a) $q(T)v = 0 \iff p_v \mid q$. In particular $p_v \mid m_T$.

(b) For a basis $v_1, \ldots, v_n$: $m_T = \operatorname{lcm}(p_{v_1}, \ldots, p_{v_n})$.

(c) If $v, Tv, \ldots, T^{n-1}v$ is a basis ($v$ is a **cyclic vector**), then $m_T = p_v$, which has degree $n$.

**The idea**

Polynomials that kill one vector behave like polynomials that kill $T$: the division argument of Prop. 65 works the same way. An operator that is 0 on a basis is 0.

Killing $T$ means killing every basis vector, i.e. being divisible by every $p_{v_i}$. The least such polynomial is the lcm.

*Proof.* 1. (a) Divide: $q = p_v s + r$ with $\deg r < \deg p_v$ or $r = 0$. If $q(T)v = 0$, then $r(T)v = q(T)v - s(T)p_v(T)v = 0$, so $r = 0$ by minimality of $p_v$. Since $m_T(T)v = 0$, $p_v \mid m_T$.

2. (b) Let $L$ be the lcm. Each $p_{v_i} \mid L$, so $L(T)v_i = 0$ by (a); $L(T)$ is 0 on a basis, so $L(T) = 0$ and $m_T \mid L$.

3. Each $p_{v_i} \mid m_T$ by (a), so $L \mid m_T$. Both are monic, so $L = m_T$.

4. (c) $v, \ldots, T^{n-1}v$ are independent, so no nonzero polynomial of degree $< n$ kills $v$: $\deg p_v \geq n$. Also $p_v \mid m_T$ and $\deg m_T \leq n$ (Prop. 67). So $m_T = p_v$.

□

**Proposition** (F8: real versus complex)

For a real matrix $A$, $m_A$ is the same over $\mathbb{R}$ and over $\mathbb{C}$. So $A$ is diagonalizable over $\mathbb{C} \iff m_A$ has distinct roots, and diagonalizable over $\mathbb{R} \iff$ in addition every root is real.

**The idea**

Def. 36 finds $m_A$ by solving linear equations whose coefficients are entries of powers of $A$, all real. Solving linear equations never leaves the real numbers.

*Proof.* 1. For each $k$, "$A^k = a_{k-1}A^{k-1} + \cdots + a_0 I$" is a system of $n^2$ linear equations in $a_0, \ldots, a_{k-1}$ with real coefficients.

2. Row-reduce the real system with real arithmetic. Whether it is solvable is read off the reduced form (no row $0 = c$ with $c \neq 0$), so the answer is the same over $\mathbb{R}$ and $\mathbb{C}$, and when it is solvable, the reduced form gives a real solution.

3. So the first solvable $k$ is the same over $\mathbb{R}$ and $\mathbb{C}$, and the solution is unique (Def. 36), hence real. So $m_A$ is the same.

4. The consequences are Prop. 68 over each field: over $\mathbb{R}$ the linear factors must have real roots.

□

**Proposition** (F9: projections onto eigenspaces, HW7 P7)

If $m_T = \prod_{i=1}^{m}(x-\lambda_i)$ with distinct roots, then $g_j(T)$, with $g_j$ as in Prop. 68, is the projection onto $E(\lambda_j, T)$ along $\bigoplus_{i\neq j} E(\lambda_i, T)$.

*Proof.* 1. Since $m_T$ has distinct roots, $V = \bigoplus_i E(\lambda_i, T)$ (Prop. 68); write $v = v_1 + \cdots + v_m$ with $v_i \in E(\lambda_i, T)$.

2. On $E(\lambda_i, T)$, $g_j(T)$ multiplies by $g_j(\lambda_i)$ (T7(b)), which is 1 if $i = j$ and 0 otherwise.

3. So $g_j(T)v = v_j$: this map fixes $E(\lambda_j, T)$ and sends $\bigoplus_{i\neq j} E(\lambda_i, T)$ to 0, which is the projection. These are the polynomials drawn in the figure after Prop. 68.

□

The **greatest common divisor** $\gcd(p, q)$ of nonzero polynomials is the monic common divisor of largest degree; the Euclidean algorithm below computes it.

**Proposition** (F10: repeated roots, Ex. 59)

Let $p$ be a nonzero polynomial.

(a) $\lambda$ is a repeated root of $p \iff p(\lambda) = p'(\lambda) = 0$.

(b) The common roots of $p$ and $p'$ are the roots of $\gcd(p, p')$, which the Euclidean algorithm finds without factoring $p$.

(c) Over $\mathbb{C}$, if $T^k = I$ for some $k \geq 1$, then $T$ is diagonalizable.

**The idea**

From calculus: at a double root the graph touches the axis, so the value and the slope are both 0.

The product rule turns this into algebra: differentiating $(x-\lambda)^2 q$ leaves a factor $x - \lambda$.

*Proof.* (a)

1. If $p = (x-\lambda)^2 q$, then $p' = 2(x-\lambda)q + (x-\lambda)^2 q'$, which is 0 at $\lambda$.

2. Conversely, if $p(\lambda) = 0$, write $p = (x-\lambda)q$ (T6). Then $p' = q + (x-\lambda)q'$, so $p'(\lambda) = q(\lambda)$. If this is 0, then $(x-\lambda) \mid q$, so $(x-\lambda)^2 \mid p$.

(b)

1. Divide $p$ by $p'$, then $p'$ by the remainder, and so on, until the remainder is 0; let $g$ be the last nonzero remainder, made monic.

2. Each remainder is (previous dividend) − (quotient)(previous divisor). So a common root of $p, p'$ is a root of every remainder, in particular of $g$.

3. Conversely, $g$ divides the last divisor, and working back up the divisions, it divides every earlier remainder, $p'$ and $p$. So the roots of $g$ are exactly the common roots of $p$ and $p'$, and $g = \gcd(p, p')$.

**Euclidean algorithm.** For $p = x^3 - 3x + 2$ and $p' = 3x^2 - 3$:

$$p = \tfrac{x}{3}\,p' + (-2x+2), \qquad p' = -\tfrac{3}{2}(x+1)\cdot(-2x+2) + 0.$$

So $g = x - 1$, and 1 is a double root: $p = (x-1)^2(x+2)$.

(c)

1. $T^k = I$ means $x^k - 1$ kills $T$, so $m_T \mid x^k - 1$.

2. $(x^k-1)' = kx^{k-1}$ has only the root 0, which is not a root of $x^k - 1$. By (a), $x^k - 1$ has no repeated root, so it is a product of $k$ distinct linear factors.

3. $m_T$ is a monic divisor of it, so it is a product of distinct linear factors (F3(b)). Prop. 68.

□

**Proposition** (F11: odd dimension, LADR 5B)

Every operator on an odd-dimensional real vector space has an eigenvalue.

**The idea**

A real eigenvalue is the same as a real root of $m_T$ (Prop. 66).

Over $\mathbb{R}$ there may be none: the rotation of $\mathbb{R}^2$ has $m = x^2 + 1$. But $\mathbb{R}^2$ has even dimension.

The proof shows that "no real root" forces the dimension to be even, in three steps. (A) A real polynomial with no real root is a product of quadratics $x^2 + bx + c$ with $b^2 < 4c$. (B) For such a quadratic $p$, $\dim \operatorname{null} p(T)$ is even: each nonzero $u$ in it comes with $Tu$, and $\operatorname{span}(u, Tu)$ is 2-dimensional. (C) Split off $\operatorname{null} p(T)$ and use induction on the dimension.

**Step A: real polynomials.** A real polynomial is a product of real linear factors and real quadratics $x^2 + bx + c$ with $b^2 < 4c$.

1. For real $p$, $p(\bar{z}) = \overline{p(z)}$, so if $z$ is a non-real root, so is $\bar{z}$.
2. The product of the two factors has real coefficients:

$$q = (x - z)(x - \bar{z}) = x^2 - 2(\operatorname{Re} z)x + |z|^2,$$

   and $b^2 - 4c = -4(\operatorname{Im} z)^2 < 0$.
3. Divide over $\mathbb{R}$: $p = qs + r$ with $\deg r < 2$. Then $r(z) = r(\bar{z}) = 0$, and a polynomial of degree $< 2$ with two roots is 0. So $p = qs$ with $s$ real. Real roots split off real linear factors (T6). Repeat on $s$ (induction on the degree).

**Step B: if $b^2 < 4c$ and $p = x^2 + bx + c$, then $\dim \operatorname{null} p(T)$ is even.** Claim, for every finite-dimensional real space $X$ and every $S \in \mathcal{L}(X)$: $\dim \operatorname{null} p(S)$ is even. Induction on $\dim \operatorname{null} p(S)$. Let $U = \operatorname{null} p(T)$, which is invariant (T5).

1. $T|_U$ has no eigenvalue: $Tu = \lambda u$ with $u \in U$, $u \neq 0$, gives $0 = p(T)u = p(\lambda)u$, but

$$p(\lambda) = (\lambda + \frac{b}{2})^2 + c - \frac{b^2}{4} > 0.$$

2. If $U = 0$, its dimension 0 is even. Otherwise pick $u \in U$, $u \neq 0$, and let $W = \operatorname{span}(u, Tu)$. It has dimension 2, since $u$ is not an eigenvector. It is invariant, since $T(Tu) = -bTu - cu$.
3. The quotient operator $S = T|_U/W$ on $U/W$ satisfies $p(S) = 0$ on $U/W$ (T10), so $\operatorname{null} p(S) = U/W$, which has dimension $\dim U - 2 < \dim U$. By the induction hypothesis (the claim for $X = U/W$) this is even, so $\dim U$ is even.

**Step C: proof of F11.** Induction on the odd number $n = \dim V$; $n = 1$ is clear.

1. Suppose $T$ has no eigenvalue. Then $m_T$ has no real root (Prop. 66), so by Step A, $m_T = ps$ with $p = x^2 + bx + c$, $b^2 < 4c$.
2. $s(T) \neq 0$ (smaller degree than $m_T$), so some $w$ has $s(T)w \neq 0$, and $p(T)\big(s(T)w\big) = 0$. So $\operatorname{null} p(T) \neq 0$, and its dimension is even by Step B.
3. $W = \operatorname{range} p(T)$ is invariant, and $\dim W = n - \dim \operatorname{null} p(T)$ is odd and less than $n$ (T1). $T|_W$ has no eigenvalue, because an eigenvector of $T|_W$ is one of $T$. This contradicts the induction hypothesis.

### Computing $m_T$

Over $\mathbb{C}$, or whenever $m_T$ splits into linear factors, $m_T = \prod_i (x - \lambda_i)^{k_i}$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues (Prop. 66), every $k_i \geq 1$, and $\sum_i k_i \leq n$ (Prop. 67). So the work is finding the eigenvalues and then the exponents. Use whichever method below needs the least computation.

**Method 1: from the definition.** Compute $I, A, A^2, \dots$ and stop at the first power that is a combination of the earlier ones. Best for $2 \times 2$ matrices or when powers are easy.

**Method 1.** $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 3 \end{smallmatrix}\right]$ is not a multiple of $I$, so $I, A$ are independent. $A^2 = \left(\begin{smallmatrix} 4 & 5 \\ 0 & 9 \end{smallmatrix}\right)$. Solve $A^2 = aA + bI$ entry by entry: entry $(1,2)$ gives $a = 5$; entry $(1,1)$ gives $4 = 2a + b$, so $b = -6$; entry $(2,2)$ checks: $9 = 3a + b$. So $m_A = x^2 - 5x + 6 = (x-2)(x-3)$.

**Method 2: eigenvalues, then test candidates.** Find the eigenvalues (for a triangular matrix, the diagonal entries, Prop. 60). The candidates are $\prod_i (x - \lambda_i)^{k_i}$ with every $k_i \geq 1$ and $\sum k_i \leq n$. Test them in increasing degree; the first one that kills $A$ is $m_A$. Reason: $m_A$ is one of the candidates, and any candidate $q$ with $q(A) = 0$ is a multiple of $m_A$ (Prop. 65), so $\deg q > \deg m_A$ unless $q = m_A$.

**Method 2, two examples.** Take

$$A = \begin{pmatrix} 3 & 1 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix},$$

which has eigenvalues $3, 3, 1$. Candidates: degree 2: $(x-3)(x-1)$; degree 3: $(x-3)^2(x-1)$ and $(x-3)(x-1)^2$.

$$(A - 3I)(A - I) = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \neq 0,$$

and $(A - 3I)^2(A - I) = 0$. So $m_A = (x-3)^2(x-1)$, and $A$ is not diagonalizable.

Next take

$$B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

which has eigenvalues $1, 1, 2$, and $(B - I)(B - 2I) = 0$. So $m_B = (x-1)(x-2)$, and $B$ is diagonalizable although it is not diagonal.

**Method 3: blocks.** For a block diagonal matrix, $m$ is the lcm of the blocks' minimal polynomials (F4). For a **block upper-triangular** matrix $A = \left[\begin{smallmatrix} B & * \\ 0 & C \end{smallmatrix}\right]$ (a matrix whose lower-left block, below the first $k$ rows and left of the last $n - k$ columns, is 0): $m_B \mid m_A$, $m_C \mid m_A$ and $m_A \mid m_B m_C$. So if $m_B$ and $m_C$ have no common root, $m_A = m_B m_C$.

**Why.**

1. The first block of basis vectors spans an invariant subspace $U$ with $B = \mathcal{M}(T|_U)$. $C$ is the matrix of $T/U$ in the cosets of the remaining basis vectors.
2. $m_B \mid m_A$ and $m_C \mid m_A$ by F3(a). $m_A \mid m_B m_C$ by step 4 of the proof of Prop. 67: $p = m_{T/U}$ sends $V$ into $U$ and $r = m_{T|_U}$ is 0 on $U$, so $r(T)p(T) = 0$.
3. "No common root" means no common root in $\mathbb{C}$. Over $\mathbb{C}$, $m_A = \prod (x - \lambda)^{e_\lambda}$, and each monic divisor of it is $\prod (x - \lambda)^{b_\lambda}$ with $b_\lambda \leq e_\lambda$ (its roots are roots of $m_A$; divide out one factor at a time, T6).
4. So $m_B$ and $m_C$ are such products over disjoint sets of $\lambda$'s, and their product divides $m_A$.

**Method 3, two examples.** Take

$$A = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}.$$

The first block $R$ has $R^2 = -I$ and is not a multiple of $I$, so $m_R = x^2 + 1$; the block $2I$ has $m = x - 2$. So $m_A = (x-2)(x^2+1)$. Over $\mathbb{C}$ the roots $2, \pm i$ are distinct, so $A$ is diagonalizable over $\mathbb{C}$; over $\mathbb{R}$ it is not, since $\pm i \notin \mathbb{R}$ (F8).

The matrix $A$ of Method 2 is $\left(\begin{smallmatrix} B & * \\ 0 & 1 \end{smallmatrix}\right)$ with $B = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, $m_B = (x-3)^2$. The roots 3 and 1 differ, so $m_A = (x-3)^2(x-1)$ with no matrix multiplication.

**Method 4: follow one vector (F7).** Compute $v, Av, A^2v, \ldots$ until the first dependence; it gives $p_v$, which divides $m_A$. If $\deg p_v = n$, then $m_A = p_v$. Otherwise repeat with more basis vectors and take the lcm.

**Method 4, two examples.** Take

$$A = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}.$$

$Ae_1 = e_2$, $Ae_2 = e_3$, so $e_1, Ae_1, A^2e_1$ is a basis. $A^3e_1 = Ae_3 = (2,-1,2) = 2e_1 - Ae_1 + 2A^2e_1$. So $m_A = x^3 - 2x^2 + x - 2 = (x-2)(x^2+1)$.

Next take

$$A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

$Ae_1 = (2,1,0)$, $A^2e_1 = (4,4,0) = 4Ae_1 - 4e_1$, so $p_{e_1} = (x-2)^2$. $Ae_3 = 3e_3$, so $p_{e_3} = x-3$. The lcm $(x-2)^2(x-3)$ has degree $3 = n$, so it is $m_A$; there is no need to compute $p_{e_2}$.

The first matrix is a **companion matrix**: if $Ae_j = e_{j+1}$ for $j < n$ and $Ae_n = a_0e_1 + \cdots + a_{n-1}e_n$, then $e_1$ is cyclic and $A^ne_1 = Ae_n$, so $m_A = x^n - a_{n-1}x^{n-1} - \cdots - a_0$. So every monic polynomial is the minimal polynomial of some matrix.

**Method 5: spot a relation.** If you notice $p(A) = 0$, then $m_A \mid p$; check which divisors work.

- Rank one, $A = uv^{\mathsf{T}}$ (all columns multiples of $u$):

  $$A^2 = u(v^{\mathsf{T}}u)v^{\mathsf{T}} = (\operatorname{tr} A)A,$$

  where $\operatorname{tr} A$ is the sum of the diagonal entries, which equals $v^{\mathsf{T}}u$. For $n \geq 2$, $A$ is not a multiple of $I$, so $m_A = x(x - \operatorname{tr} A)$ if $\operatorname{tr} A \neq 0$ (diagonalizable) and $m_A = x^2$ if $\operatorname{tr} A = 0$ (not diagonalizable). Example: $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)^2 = 2\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, so $m = x(x-2)$.
- $A^2 = A$: $m_A$ is $x$, $x-1$ or $x(x-1)$. $A^2 = I$: $m_A$ is $x-1$, $x+1$ or $(x-1)(x+1)$. Both cases are diagonalizable. $A^k = 0$: $m_A = x^j$ for the least $j$ with $A^j = 0$.
- A permutation matrix $P$ satisfies $P^k = I$ for some $k \geq 1$. There are only finitely many permutation matrices of size $n$, so two powers agree: $P^a = P^b$ with $a < b$. $P$ is invertible, so multiplying by $P^{-a}$ gives $P^{b-a} = I$. Hence $P$ is diagonalizable over $\mathbb{C}$ (F10(c)).

**Method 6: the $2\times2$ formula.** For $2\times2$ $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, with $\operatorname{tr} A = a+d$ and $\det A = ad-bc$: $m_A = x^2 - (\operatorname{tr} A)x + \det A$, unless $A = \mu I$ for a scalar $\mu$ (then $m_A = x - \mu$).

- Why: for $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, $A^2 - (a+d)A = \left(\begin{smallmatrix} bc-ad & 0 \\ 0 & bc-ad \end{smallmatrix}\right)$, so $A^2 - (\operatorname{tr} A)A + (\det A)I = 0$. If $A \neq \mu I$, $\deg m_A = 2$ (F2) and $m_A$ divides this monic quadratic, so they are equal.

**Method 6.** For $A = \left[\begin{smallmatrix} 1 & 4 \\ -1 & -3 \end{smallmatrix}\right]$: $\operatorname{tr} A = -2$ and $\det A = 1$, so $m_A = (x+1)^2$; $A \neq -I$, so $A$ is not diagonalizable.

**Method 7: operators on function spaces.** Work with the operator directly, or take its matrix in a basis.

**Method 7.** $D$ on $V = \operatorname{span}(e^x, xe^x)$: $(D-I)e^x = 0$, $(D-I)(xe^x) = e^x \neq 0$, and $(D-I)^2(xe^x) = 0$. So $(D-I)^2 = 0$ and $D \neq I$: $m_D = (x-1)^2$, and $D$ is not diagonalizable.

**Shortcuts**

- **No computation needed.** $n$ distinct eigenvalues (for example a triangular matrix with distinct diagonal entries) give $m = \prod(x - \lambda_i)$, so $T$ is diagonalizable. One eigenvalue $\lambda$ and $A \neq \lambda I$ give a non-diagonalizable $A$. Reason: every eigenvalue is a root of $m$ (Prop. 66) and $\deg m \leq n$ (Prop. 67); Prop. 68. Instance:

  $$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix} \text{ has } m = (x-1)(x-4)(x-6).$$

- **Prune candidates.** Every eigenvalue is a root of $m$ (Prop. 66), so skip any candidate that omits one. Instance: in Method 2, $(x-3)^2$ is not a candidate for $A$, since 1 is an eigenvalue.
- **Testing a candidate.** $q(A)=0$ needs every entry to be 0, so as soon as one computed entry is nonzero, the candidate fails. Instance: in Method 2, the $(1,2)$ entry of $(A-3I)(A-I)$ is 2.
- **Inverse** (F1). Instance: $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$ has $A^2-5A+6I=0$, so $A(5I-A)=6I$ and $A^{-1}=\frac{1}{6}(5I-A)$.
- **High powers.** Divide: $x^k=m(x)q(x)+r(x)$ with $\deg r<\deg m$; then $A^k=r(A)$. Find $r$ by putting each root $\lambda$ of $m$ into this identity: $r(\lambda)=\lambda^k$. For a double root, use the derivative as well: $r'(\lambda)=k\lambda^{k-1}$. (Why: $x^k-r(x)=m(x)q(x)$ is divisible by $(x-\lambda)^2$, so by F10(a) its value and its derivative are 0 at $\lambda$: $\lambda^k=r(\lambda)$ and $k\lambda^{k-1}=r'(\lambda)$.) Instances:
  - $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$, $r=ax+b$: $2a+b=2^k$, $3a+b=3^k$, so $A^k=(3^k-2^k)A+(3\cdot 2^k-2\cdot 3^k)I$.
  - $m=(x+1)^2$ (as for $\left[\begin{smallmatrix}1&4\\-1&-3\end{smallmatrix}\right]$): $r(-1)=(-1)^k$, $r'(-1)=k(-1)^{k-1}$, so $A^k=(-1)^kI+k(-1)^{k-1}(A+I)$.
- **Any polynomial** $p(A)$ equals $r(A)$, where $r$ is the remainder of $p$ divided by $m_A$. So every polynomial in $A$ is a combination of $I,A,\dots,A^{d-1}$. Reason: $p=m_Aq+r$ and $m_A(A)=0$. Instance: for $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$, $x^3=(x^2-5x+6)(x+5)+19x-30$, so $A^3=19A-30I$.
- **Eigenvectors without solving** (Prop. 66). If $m=(x-\lambda)s$, the nonzero columns of $s(A)$ are eigenvectors for $\lambda$; if $A$ is diagonalizable, they span $E(\lambda)$ (on $E(\mu)$, $s(A)$ is the number $s(\mu)$, which is 0 for $\mu\neq\lambda$ and nonzero for $\mu=\lambda$). Instance: for $A=\left[\begin{smallmatrix}2&1\\0&3\end{smallmatrix}\right]$ and $\lambda=3$,

$$s(A)=A-2I=\begin{pmatrix}0&1\\0&1\end{pmatrix},$$

  so $(1,1)$ is an eigenvector for 3.
- **Diagonalizable without finding eigenvalues.** One polynomial that kills $T$ and has distinct roots in $\mathbb{F}$ is enough, since $m_T$ divides it. Reason: the proposition on divisors of a product of linear factors, and Prop. 68. Condition: the roots must be distinct and lie in $\mathbb{F}$. Instance: $T^2=T$, $T^2=I$, $T^3=T$ (over $\mathbb{R}$), $T^k=I$ (over $\mathbb{C}$); the cyclic permutation matrix $P$ of M1(b) has $P^3=I$, so it is diagonalizable over $\mathbb{C}$ (over $\mathbb{R}$, $x^3-1$ has non-real roots, so this test says nothing).
- **Diagonalizable $\iff$ $V=\operatorname{null}(T-\lambda I)\oplus\operatorname{range}(T-\lambda I)$ for every eigenvalue** $\lambda$ (over $\mathbb{C}$). Reason: ($\Rightarrow$) in an eigenbasis, $\operatorname{null}(T-\lambda I)$ is spanned by the eigenvectors for $\lambda$ and $\operatorname{range}(T-\lambda I)$ by the others; ($\Leftarrow$) the second proof of Prop. 68 builds the eigenbasis from this splitting. Instance: for $N=\left[\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right]$, $\operatorname{null}N=\operatorname{range}N=\operatorname{span}(e_1)$, so the sum is not direct.
- **Not similar.** Different minimal polynomials prove it (F5). Equal minimal polynomials do not prove similarity. Instance: $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$ and $I$ ($m=(x-1)^2$ versus $x-1$) are not similar. $\operatorname{diag}(1,1,2)$ and $\operatorname{diag}(1,2,2)$ both have $m=(x-1)(x-2)$ but are not similar: they have different $\dim E(1)$.
- **Lower bound.** If $v,Av,\dots,A^{j-1}v$ are independent, then $\deg m_A\geq j$ (F7). Instance: Method 4, first matrix: $e_1,Ae_1,A^2e_1$ are independent, so $\deg m_A=3$.

**Traps**

- Only roots in $\mathbb{F}$ are eigenvalues: over $\mathbb{R}$, $m=x^2+1$ gives no eigenvalues.
- A polynomial that kills $T$ need not be $m_T$: $T^2=T$ allows $m_T=x$, $x-1$ or $x^2-x$. Check the divisors.
- The exponent of $x-\lambda$ in $m_T$ is not $\dim E(\lambda,T)$, in either direction. For $I_2$, $\dim E(1)=2$ but $m=x-1$ has exponent 1. For $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$, $\dim E(1)=1$ but $m=(x-1)^2$ has exponent 2.
- For a triangular matrix, $m$ depends on the entries above the diagonal: $I_2$ and $\left[\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right]$ have the same diagonal, but $m=x-1$ and $m=(x-1)^2$.
- $p(T)v=0$ for one $v\neq 0$ only gives $p_v\mid p$, not $m_T\mid p$. And $v$ need not be an eigenvector even when $p$ has distinct roots: $T=\operatorname{diag}(1,2)$, $v=(1,1)$, $p=(x-1)(x-2)$. What does follow, if $p$ splits into linear factors over $\mathbb{F}$ (always true over $\mathbb{C}$), is that some root of $p$ is an eigenvalue (T4, T6). Over $\mathbb{R}$ not even that: $T=\left[\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right]$, $p=x^2+1$, $v=e_1$.

- Over $\mathbb{R}$, $T^k = I$ does not imply diagonalizable: $R = \left[\begin{smallmatrix} 1 & -2 \\ 1 & -1 \end{smallmatrix}\right]$ has $R^4 = I$ but $m_R = x^2 + 1$.
- $m_{T^{-1}}(x)$ is not $m_T(1/x)$, which is not a polynomial. For $T = \operatorname{diag}(1, 2)$, $m_T(1/x) = (1/x - 1)(1/x - 2)$, while $m_{T^{-1}} = (x - 1)(x - \frac{1}{2})$: reverse the coefficients and divide by $c_0$.
- $\deg m_T = n$ says nothing about diagonalizability: $\operatorname{diag}(1, 2)$ (diagonalizable) and $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ (not diagonalizable) both have $\deg m = 2$.

## §3 Problem set

**HW7 P1.** Return to the matrices from Homework 6,

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

(a) Find the minimal polynomial of $A$ over $\mathbb{R}$ and over $\mathbb{C}$. (b) Find the minimal polynomial of $B$ over $\mathbb{R}$. (c) Use Prop. 68 ($T$ is diagonalizable exactly when $m_T$ is a product of distinct linear factors) to recover the conclusions of Homework 6 about whether these operators are diagonalizable over the indicated fields.

*Hint:* $A^2 = -I$ and $(B - I)^2 = 0$; neither matrix is a multiple of $I$, so $\deg m = 2$ (F2); then Prop. 68 and F8.

**HW7 P2.** Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation. (a) Find the minimal polynomial of $D$. (b) Decide whether $D$ is diagonalizable.

*Hint:* $D^4 = 0$ but $D^3 x^3 = 6$ (Ex. 54); then Prop. 68.

**HW7 P3.** Let $T$ be an invertible operator on a finite-dimensional vector space, and let $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$ be its minimal polynomial. (a) Prove that $a_0 \neq 0$. (b) Prove that $T^{-1}$ is a polynomial in $T$. (c) Suppose now that $V$ is a complex vector space and that

$$m_T(x) = x^4 + x^3 - x^2 - 5x + 4.$$

Express $T^{-1}$ explicitly as a polynomial in $T$, find the eigenvalues of $T$, and decide whether $T$ is diagonalizable.

*Hint:* (a), (b): F1. (c): $m_T(1) = m_T'(1) = 0$ (F10(a)), so divide by $(x - 1)^2$; then Props. 66 and 68.

**HW7 P4 (optional).** Let $m_T$ be the minimal polynomial of an operator $T$. (a) Prove that the minimal polynomial of $T - \lambda I$ is $m_T(x + \lambda)$. (b) If $c \neq 0$ and $d = \deg m_T$, prove that the minimal polynomial of $cT$ is $c^d m_T(x/c)$.

*Hint:* F6(a), (b): a substitution $x \mapsto x + \lambda$ or $x \mapsto x/c$ matches the polynomials that are 0 at the two operators (Prop. 65).

**HW7 P5.** Let $A$ be a $2 \times 2$ complex matrix whose only eigenvalue is $\lambda$. Prove that its minimal polynomial is either $x - \lambda$ or $(x - \lambda)^2$. Show that the first case occurs exactly when $A = \lambda I$, and conclude that $A$ is diagonalizable if and only if $A = \lambda I$.

*Hint:* Prop. 67 gives degree $\leq 2$, Prop. 66 says the only root is $\lambda$, (T6) factors over $\mathbb{C}$; then F2 and Prop. 68.

**HW7 P6.** Consider

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

Find its eigenvalues, eigenspaces, and minimal polynomial. Decide whether $C$ is diagonalizable.

*Hint:* Method 2: eigenvalues from the diagonal (Prop. 60); test $(x - 2)(x - 3)$ first.

**HW7 P7 (optional).** Let $T$ be an operator whose minimal polynomial splits as $m_T(x) = (x - \lambda_1) \cdots (x - \lambda_r)$ with distinct roots. For a fixed $j$, construct a polynomial $p_j$ such that $p_j(\lambda_j) = 1$ and $p_j(\lambda_k) = 0$ for $k \neq j$, and explain why $p_j(T)$ is the projection onto $E(\lambda_j, T)$ along the sum of the other eigenspaces.

*Hint:* the polynomials $g_j$ of Prop. 68 and F9.

**M1. Computing minimal polynomials.** For each matrix below, find the minimal polynomial and decide whether the matrix is diagonalizable (for $P$, decide over $\mathbb{R}$ and over $\mathbb{C}$).

$$\text{(a) } A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad \text{(b) } P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \text{(c) } B = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

*Hint:* (a) Method 5 (rank one); (b) $P^3 = I$ and $e_1$ is a cyclic vector (F7(c)), then F8; (c) F4 on the diagonal blocks.

**M2. Operators on polynomials.** Let $n \geq 1$. (a) Let $T\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $T(p) = xp'$. Find $m_T$ and decide whether $T$ is diagonalizable. (b) Let $S\colon \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$, $(Sp)(x) = p(x+1)$. Prove that $m_S = (x-1)^{n+1}$, and decide whether $S$ is diagonalizable.

*Hint:* (a) $T(x^k) = kx^k$. (b) $S - I$ lowers the degree by at least 1, and $(S-I)^n x^n = n!$.

**M3.** Let $V$ be a finite-dimensional real vector space and $T \in \mathcal{L}(V)$ with $T^3 = T$. Prove that $T$ is diagonalizable and that every eigenvalue of $T$ is $-1$, 0 or 1.

*Hint:* $x^3 - x = x(x-1)(x+1)$: the shortcut "Diagonalizable without finding eigenvalues", then Prop. 66.

**M4.** Let $V \neq \{0\}$ be finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $\deg m_T = 1$ if and only if $T = cI$ for some $c \in \mathbb{F}$.

*Hint:* F2.

**M5. Minimal polynomial of the inverse.** Let $T$ be invertible with $m_T(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_0$. Prove that $m_{T^{-1}}(x) = \frac{1}{a_0} x^d m_T(1/x)$.

*Hint:* F6(c).

**M6. Using the minimal polynomial to compute.** Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

Find $m_A$, write $A^{-1}$ as a polynomial in $A$, and write $A^3$ as $aA + bI$.

*Hint:* Method 6, then F1 for $A^{-1}$, then reduce $A^3$ with $A^2 = 5A + 2I$.

**R1 (Midterm 2, Question 3).** On $\mathbb{R}^3$, let $T$ have standard matrix

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(a) Determine its eigenvalues and corresponding eigenspaces. (b) Determine its minimal polynomial with justification. (c) Is $T$ diagonalizable? Explain.

*Hint:* solve the equations $(A - 2I)v = 0$ and $(A + I)v = 0$; then test both $(x-2)(x+1)$ and $(x-2)^2(x+1)$. The roots alone do not justify the exponent.

**R2 (Practice Midterm 2, Question 3).** Let $T$ act on a nonzero finite-dimensional complex space. You may use that $T$ has an eigenvalue. Prove that $T$ has an upper-triangular matrix, by induction: (a) handle dimension 1; (b) choose an eigenvalue $\lambda$, put $U = \operatorname{range}(T - \lambda I)$, and prove that $U$ is invariant and $\dim U < \dim V$; (c) obtain a basis $u_1, \ldots, u_m$ of $U$ with $Tu_j \in \operatorname{span}(u_1, \ldots, u_j)$; (d) extend it by $v_1, \ldots, v_r$ and use $Tv_j = (T - \lambda I)v_j + \lambda v_j$.

*Hint:* Prop. 61; include the case $U = \{0\}$.

**R3 (Practice Midterm 2, Question 4).** Let $T$ act on a nonzero finite-dimensional real space. (a) If $m_T(0) \neq 0$, prove that $T$ is invertible by expressing $T^{-1}$ as a polynomial in $T$. (b) If $m_T(0) = 0$, prove that $T$ is not invertible, using minimality. (c) For $R(x, y) = (-y, x)$ on $\mathbb{R}^2$, find $m_R$. (d) Use (a) to express $R^{-1}$ in terms of $R$ and check directly.

*Hint:* factor $T$ out of every nonconstant term; in (b), an inverse would cancel a factor $x$ and give a relation of smaller degree.

**R4 (Midterm 2 review, Problem 4).** (a) On the real space $V = \operatorname{span}(1, \sin x, \cos x, \sin 2x, \cos 2x)$, find the eigenvalues of differentiation and decide whether it is diagonalizable. (b) For $T(x, y) = (y, -x)$ on $\mathbb{C}^2$, find the eigenvalues and a basis giving an upper-triangular matrix. (c) Show that the same formula on $\mathbb{R}^2$ has no upper-triangular basis.

*Hint:* (a) write $Df = \lambda f$ in coordinates for the five spanning functions; (b), (c): $T^2 = -I$, and an upper-triangular basis starts with an eigenvector.

**R5 (Midterm 2 review, Problem 6; LADR 5C, Exercise 7).** Let $T \in \mathcal{L}(V)$ and $v \in V$. Prove that there is a unique monic polynomial $p_v$ of least degree with $p_v(T)v = 0$, and that $p_v$ divides $m_T$.

*Hint:* F7(a); for uniqueness, two such monic polynomials differ by one of smaller degree that is 0 on $v$.

**R6 (Midterm 2 review, Problem 8: true or false).** (a) Every vector space can be written as a direct sum of eigenspaces of some operator. (b) An operator on a real vector space always has an eigenvector. (c) The dimension of $E(\lambda, T)$ is the multiplicity of $x - \lambda$ in the minimal polynomial. (d) For an upper-triangular matrix, the minimal polynomial depends on the entries above the diagonal.

*Hint:* each statement has a $2 \times 2$ or $3 \times 3$ example in the Traps boxes; for (b) also see F11.

**R7 (Additional Midterm Practice).** Let $T \in \mathcal{L}(V)$ with $T^2 = T$. (a) Prove that $\operatorname{range} T \cap \operatorname{null} T = \{0\}$. (b) Prove that $V = \operatorname{range} T + \operatorname{null} T$. (c) If $S \in \mathcal{L}(V)$ and $ST = TS$, prove that $\operatorname{range} T$ and $\operatorname{null} T$ are invariant under $S$, so that $S$ has a block diagonal matrix in a basis of $\operatorname{range} T$ followed by a basis of $\operatorname{null} T$.

*Hint:* (a), (b): write $v = Tv + (v - Tv)$; (c): the argument of (T5) with $S$ in place of $p(T)$, then (T8).